

```
import torch

print("PyTorch:", torch.__version__)
print("CUDA Available:", torch.cuda.is_available())

if torch.cuda.is_available():
    print("GPU:", torch.cuda.get_device_name(0))
```

PyTorch: 2.11.0+cu128
 CUDA Available: True
 GPU: Tesla T4

```
!pip install -q ultralytics
```

```

  _____ 41.3/41.3 kB 3.6 MB/s eta 0:00:00
  _____ 1.3/1.3 MB 57.4 MB/s eta 0:00:00
```

```
from ultralytics import YOLO

print("Ultralytics installed successfully")
```

Creating new Ultralytics Settings v0.0.6 file 
 View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
 Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics>
 Ultralytics installed successfully

```
!ls /content
```

```
sample_data  --v2.v7-add-test-potato-in-internet.yolov8.zip
```

```
!file "/content/--v2.v7-add-test-potato-in-internet.yolov8.zip"
```

```
/content/--v2.v7-add-test-potato-in-internet.yolov8.zip: Zip archive data, at least v2.0 to extract, compression method=deflate
```

```
!ls -lh "/content/--v2.v7-add-test-potato-in-internet.yolov8.zip"
```

```
-rw-r--r-- 1 root root 76M Jun  9 12:19 /content/--v2.v7-add-test-potato-in-internet.yolov8.zip
```

```
!unzip -q "/content/--v2.v7-add-test-potato-in-internet.yolov8.zip" -d /content/dataset
```

```
!find /content/dataset -maxdepth 2 -type d
```

```

/content/dataset
/content/dataset/train
/content/dataset/train/images
/content/dataset/train/labels
/content/dataset/valid
/content/dataset/valid/images
/content/dataset/valid/labels
/content/dataset/test
/content/dataset/test/images
/content/dataset/test/labels
```

```
!find /content/dataset -name "*.yaml"
```

```
/content/dataset/data.yaml
```

```
!cat /content/dataset/data.yaml
```

```

train: ../train/images
val: ../valid/images
test: ../test/images

nc: 4
names: ['apple', 'banana', 'orange', 'potato']

roboflow:
  workspace: mine-uyi8t
  project: -v2
  version: 7
  license: CC BY 4.0
  url: https://universe.roboflow.com/mine-uyi8t/-v2/dataset/7
```

```
import os

print(os.path.exists("/content/dataset/train/images"))
print(os.path.exists("/content/dataset/valid/images"))
print(os.path.exists("/content/dataset/test/images"))

True
True
True
```

```
from ultralytics import YOLO

model = YOLO("yolov8n.pt")

results = model.train(
    data="/content/dataset/data.yaml",
    epochs=1,
    imgsz=640,
    batch=16,
    device=0,
    project="fruit_detection",
    name="sanity_test"
)
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt> to 'yolov8n.pt': 100% 6.2MB
Ultralytics 8.4.62 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5,
Downloading <https://ultralytics.com/assets/Arial.ttf> to '/root/.config/Ultralytics/Arial.ttf': 100% 755.1KB 101.4M
Overriding model.yaml nc=80 with nc=4

	from	n	params	module	arguments	
0		-1	1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1		-1	1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2		-1	1	7360	ultralytics.nn.modules.block.C2f	[32, 32, 1, True]
3		-1	1	18560	ultralytics.nn.modules.conv.Conv	[32, 64, 3, 2]
4		-1	2	49664	ultralytics.nn.modules.block.C2f	[64, 64, 2, True]
5		-1	1	73984	ultralytics.nn.modules.conv.Conv	[64, 128, 3, 2]
6		-1	2	197632	ultralytics.nn.modules.block.C2f	[128, 128, 2, True]
7		-1	1	295424	ultralytics.nn.modules.conv.Conv	[128, 256, 3, 2]
8		-1	1	460288	ultralytics.nn.modules.block.C2f	[256, 256, 1, True]
9		-1	1	164608	ultralytics.nn.modules.block.SPPF	[256, 256, 5]
10		-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
11	[-1, 6]		1	0	ultralytics.nn.modules.conv.Concat	[1]
12		-1	1	148224	ultralytics.nn.modules.block.C2f	[384, 128, 1]
13		-1	1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
14	[-1, 4]		1	0	ultralytics.nn.modules.conv.Concat	[1]
15		-1	1	37248	ultralytics.nn.modules.block.C2f	[192, 64, 1]
16		-1	1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
17	[-1, 12]		1	0	ultralytics.nn.modules.conv.Concat	[1]
18		-1	1	123648	ultralytics.nn.modules.block.C2f	[192, 128, 1]
19		-1	1	147712	ultralytics.nn.modules.conv.Conv	[128, 128, 3, 2]
20	[-1, 9]		1	0	ultralytics.nn.modules.conv.Concat	[1]
21		-1	1	493056	ultralytics.nn.modules.block.C2f	[384, 256, 1]
22	[15, 18, 21]		1	752092	ultralytics.nn.modules.head.Detect	[4, 16, None, [64, 128, 256]]

Model summary: 130 layers, 3,011,628 parameters, 3,011,612 gradients, 8.2 GFLOPs

Transferred 319/355 items from pretrained weights
Freezing layer 'model.22.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt> to 'yolo26n.pt': 100% 5.3MB
AMP: checks passed
train: Fast image access (ping: 0.0±0.0 ms, read: 862.6±298.0 MB/s, size: 24.8 KB)
train: Scanning /content/dataset/train/labels... 1852 images, 0 backgrounds, 0 corrupt: 100% 1852/1852 2.4Kit/s 0
train: New cache created: /content/dataset/train/labels.cache
augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average')
val: Fast image access (ping: 0.0±0.1 ms, read: 553.6±228.9 MB/s, size: 32.1 KB)
val: Scanning /content/dataset/valid/labels... 266 images, 0 backgrounds, 0 corrupt: 100% 266/266 1.5Kit/s 0.2s
val: New cache created: /content/dataset/valid/labels.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum'
optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 57 weight(decay=0.0), 64 weight(decay=0.0005), 63 bias(decay=0.0)
Plotting labels to /content/runs/detect/fruit_detection/sanity_test/labels.jpg...
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to /content/runs/detect/fruit_detection/sanity_test
Starting training for 1 epochs...

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size
1/1	2G	0.4815	1.557	1.087	12	640: 100% 116/116 3.7it/s 31.6s
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100% 9/9 1.8it/s 4.9s
	all	266	280	0.849	0.726	0.932 0.816

```
from ultralytics import YOLO

model = YOLO("yolov10n.pt")
print("YOLOv10 loaded successfully")
```

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov10n.pt> to 'yolov10n.pt': 100% ————— 5.6MB
YOLOv10 loaded successfully

```
from ultralytics import YOLO

model = YOLO("yolov10n.pt")

results_10 = model.train(
    data="/content/dataset/data.yaml",
    epochs=10,
    imgsz=640,
    batch=16,
    device=0,
    project="YOLOv10_Fruit_Project",
    name="epoch_10"
)
```

Ultralytics 8.4.62 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5,
Overriding model.yaml nc=80 with nc=4

	from	n	params	module	arguments
0		-1 1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1		-1 1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2		-1 1	7360	ultralytics.nn.modules.block.C2f	[32, 32, 1, True]
3		-1 1	18560	ultralytics.nn.modules.conv.Conv	[32, 64, 3, 2]
4		-1 2	49664	ultralytics.nn.modules.block.C2f	[64, 64, 2, True]
5		-1 1	9856	ultralytics.nn.modules.block.SCDown	[64, 128, 3, 2]
6		-1 2	197632	ultralytics.nn.modules.block.C2f	[128, 128, 2, True]
7		-1 1	36096	ultralytics.nn.modules.block.SCDown	[128, 256, 3, 2]
8		-1 1	460288	ultralytics.nn.modules.block.C2f	[256, 256, 1, True]
9		-1 1	164608	ultralytics.nn.modules.block.SPPF	[256, 256, 5]
10		-1 1	249728	ultralytics.nn.modules.block.PSA	[256, 256]
11		-1 1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
12	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	[1]
13		-1 1	148224	ultralytics.nn.modules.block.C2f	[384, 128, 1]
14		-1 1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
15	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat	[1]
16		-1 1	37248	ultralytics.nn.modules.block.C2f	[192, 64, 1]
17		-1 1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
18	[-1, 13]	1	0	ultralytics.nn.modules.conv.Concat	[1]
19		-1 1	123648	ultralytics.nn.modules.block.C2f	[192, 128, 1]
20		-1 1	18048	ultralytics.nn.modules.block.SCDown	[128, 128, 3, 2]
21	[-1, 10]	1	0	ultralytics.nn.modules.conv.Concat	[1]
22		-1 1	282624	ultralytics.nn.modules.block.C2fCIB	[384, 256, 1, True, True]
23	[16, 19, 22]	1	862888	ultralytics.nn.modules.head.v10Detect	[4, [64, 128, 256]]

YOLOv10n summary: 224 layers, 2,708,600 parameters, 2,708,584 gradients, 8.4 GFLOPs

Transferred 493/595 items from pretrained weights
Freezing layer 'model.23.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access (ping: 0.0±0.0 ms, read: 1391.5±511.9 MB/s, size: 30.2 KB)
train: Scanning /content/dataset/train/labels.cache... 1852 images, 0 backgrounds, 0 corrupt: 100% ————— 1852/1852 388.4
augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average')
val: Fast image access (ping: 0.0±0.0 ms, read: 333.6±132.6 MB/s, size: 29.4 KB)
val: Scanning /content/dataset/valid/labels.cache... 266 images, 0 backgrounds, 0 corrupt: 100% ————— 266/266 15.3Mit/s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum'
optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 95 weight(decay=0.0), 108 weight(decay=0.0005), 107 bias(decay=0.0)
Plotting labels to /content/runs/detect/YOLOv10_Fruit_Project/epoch_10/labels.jpg...
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to /content/runs/detect/YOLOv10_Fruit_Project/epoch_10
Starting training for 10 epochs...
Closing dataloader mosaic
augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average')

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
1/10	2.81G	0.348	8.216	0.9967	12	640: 100%	116/116 2.1it/s 55.8s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 9/9 1.1it/s 8.4s
	all	266	280	0.74	0.564	0.746	0.688
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
2/10	3.27G	0.3867	5.231	1.012	12	640: 100%	116/116 3.5it/s 33.1s

```
!ls /content/runs/detect/YOLOv10_Fruit_Project/epoch_10/weights
```

```
best.pt  last.pt
```

```
!tail -20 /content/runs/detect/YOLOv10_Fruit_Project/epoch_10/results.csv
```

```
epoch,time,train/box_loss,train/cls_loss,train/df1_loss,metrics/precision(B),metrics/recall(B),metrics/mAP50(B),metrics/mAP50-95
1,64.6903,0.34801,8.21612,0.99671,0.73964,0.56444,0.74575,0.68771,0.38518,8.77051,1.02457,0.000413075,0.000413075,0.000413075
2,100.193,0.3867,5.23116,1.01226,0.81501,0.79231,0.85534,0.76385,0.43065,4.2902,1.10449,0.000747597,0.000747597,0.000747597
3,135.121,0.39021,2.88433,1.0039,0.82072,0.79867,0.90845,0.81779,0.42557,2.30612,1.02616,0.000999619,0.000999619,0.000999619
4,169.313,0.36442,1.45829,0.98731,0.84054,0.93901,0.95055,0.88214,0.36953,1.14993,0.98088,0.00087875,0.00087875,0.00087875
5,204.069,0.33797,0.8543,0.96288,0.9236,0.90783,0.97514,0.91666,0.3522,0.71812,0.94346,0.000755,0.000755,0.000755
6,237.417,0.30709,0.59201,0.93514,0.80886,0.89903,0.90454,0.86123,0.30273,0.79514,0.91178,0.00063125,0.00063125,0.00063125
7,271.552,0.27607,0.44911,0.91546,0.97972,0.90856,0.97951,0.94999,0.28822,0.4208,0.89206,0.0005075,0.0005075,0.0005075
8,306.175,0.26057,0.33945,0.90088,0.9817,0.96203,0.98303,0.957,0.26429,0.32529,0.88552,0.00038375,0.00038375,0.00038375
9,339.606,0.23665,0.28255,0.88358,0.95597,0.9686,0.98248,0.96146,0.24323,0.32195,0.8712,0.00026,0.00026,0.00026
10,374.679,0.2166,0.26622,0.87018,0.9534,0.97231,0.98293,0.9642,0.2352,0.30192,0.86683,0.00013625,0.00013625,0.00013625
```

```
from google.colab import files
```

```
files.download("/content/runs/detect/YOLOv10_Fruit_Project/epoch_10/weights/best.pt")
```

```
!zip -r epoch10_results.zip /content/runs/detect/YOLOv10_Fruit_Project/epoch_10
```

```
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/ (stored 0%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/val_batch0_labels.jpg (deflated 13%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/confusion_matrix.png (deflated 29%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/labels.jpg (deflated 47%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/val_batch1_labels.jpg (deflated 17%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/BoxR_curve.png (deflated 13%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/args.yaml (deflated 52%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/BoxF1_curve.png (deflated 12%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/BoxP_curve.png (deflated 15%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/val_batch2_pred.jpg (deflated 10%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/train_batch1.jpg (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/confusion_matrix_normalized.png (deflated 27%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/BoxPR_curve.png (deflated 22%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/val_batch1_pred.jpg (deflated 16%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/val_batch2_labels.jpg (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/results.png (deflated 7%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/weights/ (stored 0%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/weights/last.pt (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/weights/best.pt (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/train_batch0.jpg (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/results.csv (deflated 57%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/train_batch2.jpg (deflated 12%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_10/val_batch0_pred.jpg (deflated 12%)
```

```
from google.colab import files
```

```
files.download("epoch10_results.zip")
```

```
from ultralytics import YOLO
```

```
model = YOLO("yolov10n.pt")
```

```
results_20 = model.train(
    data="/content/dataset/data.yaml",
    epochs=20,
    imgsz=640,
    batch=16,
    device=0,
    project="YOLOv10_Fruit_Project",
    name="epoch_20"
)
```

```
Ultralytics 8.4.62 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5,
Overriding model.yaml nc=80 with nc=4
```

		from	n	params	module	arguments
0		-1	1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1		-1	1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]

```

2         -1 1 7360 ultralytics.nn.modules.block.C2f [32, 32, 1, True]
3         -1 1 18560 ultralytics.nn.modules.conv.Conv [32, 64, 3, 2]
4         -1 2 49664 ultralytics.nn.modules.block.C2f [64, 64, 2, True]
5         -1 1 9856 ultralytics.nn.modules.block.SCDow [64, 128, 3, 2]
6         -1 2 197632 ultralytics.nn.modules.block.C2f [128, 128, 2, True]
7         -1 1 36096 ultralytics.nn.modules.block.SCDow [128, 256, 3, 2]
8         -1 1 460288 ultralytics.nn.modules.block.C2f [256, 256, 1, True]
9         -1 1 164608 ultralytics.nn.modules.block.SPPF [256, 256, 5]
10        -1 1 249728 ultralytics.nn.modules.block.PSA [256, 256]
11        -1 1 0 torch.nn.modules.upsampling.Upsample [None, 2, 'nearest']
12        [-1, 6] 1 0 ultralytics.nn.modules.conv.Concat [1]
13        -1 1 148224 ultralytics.nn.modules.block.C2f [384, 128, 1]
14        -1 1 0 torch.nn.modules.upsampling.Upsample [None, 2, 'nearest']
15        [-1, 4] 1 0 ultralytics.nn.modules.conv.Concat [1]
16        -1 1 37248 ultralytics.nn.modules.block.C2f [192, 64, 1]
17        -1 1 36992 ultralytics.nn.modules.conv.Conv [64, 64, 3, 2]
18        [-1, 13] 1 0 ultralytics.nn.modules.conv.Concat [1]
19        -1 1 123648 ultralytics.nn.modules.block.C2f [192, 128, 1]
20        -1 1 18048 ultralytics.nn.modules.block.SCDow [128, 128, 3, 2]
21        [-1, 10] 1 0 ultralytics.nn.modules.conv.Concat [1]
22        -1 1 282624 ultralytics.nn.modules.block.C2fCIB [384, 256, 1, True, True]
23        [16, 19, 22] 1 862888 ultralytics.nn.modules.head.v10Detect [4, [64, 128, 256]]
YOLOv10n summary: 224 layers, 2,708,600 parameters, 2,708,584 gradients, 8.4 GFLOPs

```

Transferred 493/595 items from pretrained weights

Freezing layer 'model.23.dfl.conv.weight'

AMP: running Automatic Mixed Precision (AMP) checks...

AMP: checks passed 

train: Fast image access  (ping: 0.0±0.0 ms, read: 1084.2±397.3 MB/s, size: 30.2 KB)

train: Scanning /content/dataset/train/labels.cache... 1852 images, 0 backgrounds, 0 corrupt: 100%  1852/1852 431.

augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average')

val: Fast image access  (ping: 0.0±0.0 ms, read: 303.8±220.5 MB/s, size: 29.4 KB)

val: Scanning /content/dataset/valid/labels.cache... 266 images, 0 backgrounds, 0 corrupt: 100%  266/266 13.3Mit/s

optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum'

optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 95 weight(decay=0.0), 108 weight(decay=0.0005), 107 bias(decay=0.0)

Plotting labels to /content/runs/detect/YOLOv10_Fruit_Project/epoch_20/labels.jpg...

Image sizes 640 train, 640 val

Using 2 dataloader workers

Logging results to /content/runs/detect/YOLOv10_Fruit_Project/epoch_20

Starting training for 20 epochs...

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	Speed
1/20	2.83G	0.4792	4.403	1.083	12	640: 100%  116/116 3.2it/s 36.6s	
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100%  9/9 4.4it/s 2.1s	
	all	266	280	0.758	0.645	0.792 0.74	
2/20	3.29G	0.5567	2.815	1.094	12	640: 100%  116/116 3.4it/s 34.2s	
	Class	Images	Instances	Box(P	R	mAP50 mAP50-95): 100%  9/9 3.3it/s 2.8s	
	all	266	280	0.763	0.745	0.785 0.699	

from google.colab import files

files.download("/content/runs/detect/YOLOv10_Fruit_Project/epoch_20/weights/best.pt")

!zip -r epoch20_results.zip /content/runs/detect/YOLOv10_Fruit_Project/epoch_20

```

adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/ (stored 0%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/val_batch0_labels.jpg (deflated 13%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/confusion_matrix.png (deflated 29%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/labels.jpg (deflated 47%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/val_batch1_labels.jpg (deflated 17%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/BoxR_curve.png (deflated 15%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/args.yaml (deflated 52%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/BoxF1_curve.png (deflated 14%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/BoxP_curve.png (deflated 19%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/val_batch2_pred.jpg (deflated 10%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/train_batch1160.jpg (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/train_batch1.jpg (deflated 6%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/train_batch1161.jpg (deflated 12%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/confusion_matrix_normalized.png (deflated 27%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/BoxPR_curve.png (deflated 24%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/val_batch1_pred.jpg (deflated 16%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/val_batch2_labels.jpg (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/results.png (deflated 8%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/weights/ (stored 0%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/weights/last.pt (deflated 10%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/weights/best.pt (deflated 10%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/train_batch0.jpg (deflated 5%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/results.csv (deflated 60%)

```

adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/train_batch2.jpg (deflated 4%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/train_batch1162.jpg (deflated 9%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/val_batch0_pred.jpg (deflated 12%)

```
from google.colab import files

files.download("epoch20_results.zip")
```

```
from ultralytics import YOLO

model = YOLO("yolov10n.pt")

results_30 = model.train(
    data="/content/dataset/data.yaml",
    epochs=30,
    imgsz=640,
    batch=16,
    device=0,
    project="YOLOv10_Fruit_Project",
    name="epoch_30"
)
```

Ultralytics 8.4.62 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5, (Overriding model.yaml nc=80 with nc=4)

	from	n	params	module	arguments
0		-1 1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1		-1 1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2		-1 1	7360	ultralytics.nn.modules.block.C2f	[32, 32, 1, True]
3		-1 1	18560	ultralytics.nn.modules.conv.Conv	[32, 64, 3, 2]
4		-1 2	49664	ultralytics.nn.modules.block.C2f	[64, 64, 2, True]
5		-1 1	9856	ultralytics.nn.modules.block.SCDown	[64, 128, 3, 2]
6		-1 2	197632	ultralytics.nn.modules.block.C2f	[128, 128, 2, True]
7		-1 1	36096	ultralytics.nn.modules.block.SCDown	[128, 256, 3, 2]
8		-1 1	460288	ultralytics.nn.modules.block.C2f	[256, 256, 1, True]
9		-1 1	164608	ultralytics.nn.modules.block.SPPF	[256, 256, 5]
10		-1 1	249728	ultralytics.nn.modules.block.PSA	[256, 256]
11		-1 1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
12	[-1, 6]	1	0	ultralytics.nn.modules.conv.Concat	[1]
13		-1 1	148224	ultralytics.nn.modules.block.C2f	[384, 128, 1]
14		-1 1	0	torch.nn.modules.upsampling.Upsample	[None, 2, 'nearest']
15	[-1, 4]	1	0	ultralytics.nn.modules.conv.Concat	[1]
16		-1 1	37248	ultralytics.nn.modules.block.C2f	[192, 64, 1]
17		-1 1	36992	ultralytics.nn.modules.conv.Conv	[64, 64, 3, 2]
18	[-1, 13]	1	0	ultralytics.nn.modules.conv.Concat	[1]
19		-1 1	123648	ultralytics.nn.modules.block.C2f	[192, 128, 1]
20		-1 1	18048	ultralytics.nn.modules.block.SCDown	[128, 128, 3, 2]
21	[-1, 10]	1	0	ultralytics.nn.modules.conv.Concat	[1]
22		-1 1	282624	ultralytics.nn.modules.block.C2fCIB	[384, 256, 1, True, True]
23	[16, 19, 22]	1	862888	ultralytics.nn.modules.head.v10Detect	[4, [64, 128, 256]]

YOLOv10n summary: 224 layers, 2,708,600 parameters, 2,708,584 gradients, 8.4 GFLOPs

Transferred 493/595 items from pretrained weights
Freezing layer 'model.23.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: checks passed
train: Fast image access (ping: 0.0±0.0 ms, read: 1322.2±545.2 MB/s, size: 30.2 KB)
train: Scanning /content/dataset/train/labels.cache... 1852 images, 0 backgrounds, 0 corrupt: 100% 1852/1852 554.1
augmentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average')
val: Fast image access (ping: 0.0±0.0 ms, read: 312.1±186.7 MB/s, size: 29.4 KB)
val: Scanning /content/dataset/valid/labels.cache... 266 images, 0 backgrounds, 0 corrupt: 100% 266/266 11.3Mit/s
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and 'momentum=0.937' and determining best 'optimizer', 'lr0' and 'momentum'
optimizer: AdamW(lr=0.00125, momentum=0.9) with parameter groups 95 weight(decay=0.0), 108 weight(decay=0.0005), 107 bias(decay=0.0)
Plotting labels to /content/runs/detect/YOLOv10_Fruit_Project/epoch_30/labels.jpg...
Image sizes 640 train, 640 val
Using 2 dataloader workers
Logging results to /content/runs/detect/YOLOv10_Fruit_Project/epoch_30
Starting training for 30 epochs...

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
1/30	2.83G	0.4792	4.403	1.083	12	640: 100%	116/116 3.1it/s 37.8s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 9/9 4.5it/s 2.0s
	all	266	280	0.758	0.645	0.792	0.74
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
2/30	3.29G	0.5492	2.811	1.085	12	640: 100%	116/116 3.1it/s 36.8s
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100% 9/9 4.8it/s 1.9s
	all	266	280	0.805	0.641	0.799	0.721


```
from google.colab import files
```

```
files.download("/content/runs/detect/YOLOv10_Fruit_Project/epoch_30/weights/best.pt")
```

```
!zip -r epoch30_results.zip /content/runs/detect/YOLOv10_Fruit_Project/epoch_30
```

```
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/ (stored 0%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch0_labels.jpg (deflated 13%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/confusion_matrix.png (deflated 29%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/labels.jpg (deflated 47%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch2320.jpg (deflated 9%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch1_labels.jpg (deflated 17%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/BoxR_curve.png (deflated 14%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/args.yaml (deflated 52%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/BoxF1_curve.png (deflated 13%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/BoxP_curve.png (deflated 18%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch2_pred.jpg (deflated 10%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch1.jpg (deflated 6%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/confusion_matrix_normalized.png (deflated 28%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/BoxPR_curve.png (deflated 23%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch1_pred.jpg (deflated 16%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch2_labels.jpg (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/results.png (deflated 7%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/weights/ (stored 0%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/weights/last.pt (deflated 10%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/weights/best.pt (deflated 10%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch2322.jpg (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch0.jpg (deflated 5%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch2321.jpg (deflated 8%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/results.csv (deflated 61%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch2.jpg (deflated 4%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch0_pred.jpg (deflated 12%)
```

```
from google.colab import files
```

```
files.download("epoch30_results.zip")
```

```
!zip -r YOLOv10_Project_Complete.zip /content/runs
```

```
adding: content/runs/ (stored 0%)
adding: content/runs/detect/ (stored 0%)
adding: content/runs/detect/fruit_detection/ (stored 0%)
adding: content/runs/detect/fruit_detection/sanity_test/ (stored 0%)
adding: content/runs/detect/fruit_detection/sanity_test/val_batch0_labels.jpg (deflated 13%)
adding: content/runs/detect/fruit_detection/sanity_test/confusion_matrix.png (deflated 27%)
adding: content/runs/detect/fruit_detection/sanity_test/labels.jpg (deflated 47%)
adding: content/runs/detect/fruit_detection/sanity_test/val_batch1_labels.jpg (deflated 17%)
adding: content/runs/detect/fruit_detection/sanity_test/BoxR_curve.png (deflated 8%)
adding: content/runs/detect/fruit_detection/sanity_test/args.yaml (deflated 53%)
adding: content/runs/detect/fruit_detection/sanity_test/BoxF1_curve.png (deflated 7%)
adding: content/runs/detect/fruit_detection/sanity_test/BoxP_curve.png (deflated 11%)
adding: content/runs/detect/fruit_detection/sanity_test/val_batch2_pred.jpg (deflated 10%)
adding: content/runs/detect/fruit_detection/sanity_test/train_batch1.jpg (deflated 6%)
adding: content/runs/detect/fruit_detection/sanity_test/confusion_matrix_normalized.png (deflated 26%)
adding: content/runs/detect/fruit_detection/sanity_test/BoxPR_curve.png (deflated 17%)
adding: content/runs/detect/fruit_detection/sanity_test/val_batch1_pred.jpg (deflated 16%)
adding: content/runs/detect/fruit_detection/sanity_test/val_batch2_labels.jpg (deflated 11%)
adding: content/runs/detect/fruit_detection/sanity_test/results.png (deflated 22%)
adding: content/runs/detect/fruit_detection/sanity_test/weights/ (stored 0%)
adding: content/runs/detect/fruit_detection/sanity_test/weights/last.pt (deflated 9%)
adding: content/runs/detect/fruit_detection/sanity_test/weights/best.pt (deflated 9%)
adding: content/runs/detect/fruit_detection/sanity_test/train_batch0.jpg (deflated 5%)
adding: content/runs/detect/fruit_detection/sanity_test/results.csv (deflated 47%)
adding: content/runs/detect/fruit_detection/sanity_test/train_batch2.jpg (deflated 4%)
adding: content/runs/detect/fruit_detection/sanity_test/val_batch0_pred.jpg (deflated 12%)
adding: content/runs/detect/YOLOv10_Fruit_Project/ (stored 0%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/ (stored 0%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch0_labels.jpg (deflated 13%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/confusion_matrix.png (deflated 29%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/labels.jpg (deflated 47%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch2320.jpg (deflated 9%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch1_labels.jpg (deflated 17%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/BoxR_curve.png (deflated 14%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/args.yaml (deflated 52%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/BoxF1_curve.png (deflated 13%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/BoxP_curve.png (deflated 18%)
```

```
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch2_pred.jpg (deflated 10%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch1.jpg (deflated 6%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/confusion_matrix_normalized.png (deflated 28%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/BoxPR_curve.png (deflated 23%)
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adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch2_labels.jpg (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/results.png (deflated 7%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/weights/ (stored 0%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/weights/last.pt (deflated 10%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/weights/best.pt (deflated 10%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch2322.jpg (deflated 11%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch0.jpg (deflated 5%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch2321.jpg (deflated 8%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/results.csv (deflated 61%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/train_batch2.jpg (deflated 4%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_30/val_batch0_pred.jpg (deflated 12%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/ (stored 0%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/val_batch0_labels.jpg (deflated 13%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/confusion_matrix.png (deflated 29%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/labels.jpg (deflated 47%)
adding: content/runs/detect/YOLOv10_Fruit_Project/epoch_20/val_batch1_labels.jpg (deflated 17%)
```

```
from google.colab import files
files.download("YOLOv10_Project_Complete.zip")
```